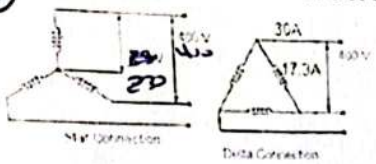
Answer the following questions by circling the right answer in the table provided at the end of this question

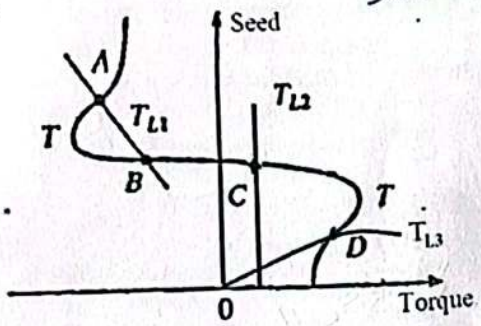
1. for the given figures of ac motor stator connections, if we start the motor using Y –connection, then the line current will be

a. 17.23A b. 51A c. 1.73A d. 0A

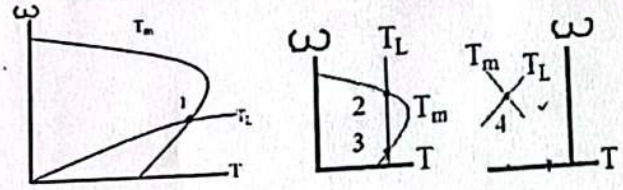


2.Stabilities for points A,B,C,D respectively in the figure below are

a. unstable, stable, stable, stable
b. unstable, unstable, stable, unstable
c. unstable, unstable, stable, unstable
d. unstable, stable, stable, unstable .



3.Unstable points of equilibrium are



a. 1 and 3 b. 3 only c. 2 and 3 d. 3 and 4

4.If the torque developed by the motor is $T = T_L \pm J\frac{d\omega}{dt}$, the drive is

a. decelerating b. accelerating c. working at transient condition d. working at S.S.P.

5. In the electric drive system, the main function of the power converter is

a. matching the motor and MTD to meet the load requirements
b. adapting to the changes of the environment only
c. sensing the feedback variables d. providing the motor with the suitable voltage signal

6. If the two loads are driven by one motor, the drive classification is

a. interconnected b. regulated c. auxiliary d. group

7. The electric drive system with following motor torque and load torque expressions is stable:

a. $T_m=10+N, T_L=5$ b. $T_m=10+\omega, T_L=5$
c. $T_m=10-\omega, T_L=10$ d. $T_m=25+5\omega, T_L=2+2\omega^2$

8. The drive system input power is:

a. Thermal power b. Electrical power c. Mechanical power d. Electromagnetic power

9. Component determines the motor starting time is

a. motor inertia b. motor size c. speed range d. all of these

10. Torque equation for rotational motion when inertia is variable

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a. $F_m - F_f = \frac{d(Mv)}{dt}$

b. $T - T_L = J\frac{d\omega_m}{dt} - \omega_m \frac{dJ}{dt}$

c. $T - T_L = J\frac{d\omega_m}{dt}$

d. $T - T_L = J\frac{d^2\theta}{dt^2}$

11. The active load torque industrial machinery is:
a. pump b. compressor c. escalator d. turning machine

12. Rotary vane compressor torque-speed characteristic is:
a. decreasing linear characteristic b. independent of speed
c. decreasing non-linear characteristic d. quadratic characteristic

13. In drive systems, problems of fluctuating loads are overcome by
a. mounting flywheels on the motor shaft in non-reversible drive b. using step-down transformers
c. using voltage stabilizers d. using capacitive compensators

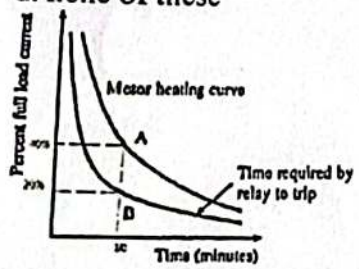
14. Crane motor lowers load of 10 N.m through gearbox, which has gear ratio of 5. If the system efficiency equals 0.8, the referred load torque to motor shaft is:
a. 2.5N.m b. 1.6N.m c. 40N.m d. 1.8N.m

15. Electric motor has moment of inertia of 2kg.m², connected through gearbox of 10 gear ratio to drum with moment of inertia 4 kg.m². The drive system total (equivalent) moment of inertia equals:
a. 4.02 kg.m² b. 0.06 kg.m² c. 2.04 kg.m² d. 2.4N.m²

16. In four quadrant operation of a motor driving a hoist load(elevator), if the cage is empty and the motor speed is downward, then the motor is working in
a. the first quadrant b. the second quadrant c. the third quadrant d. the fourth quadrant

17. If the motor works in the second quadrant. This mode of operation is a
a. reverse braking b. reverse motoring c. motoring mode d. forward braking

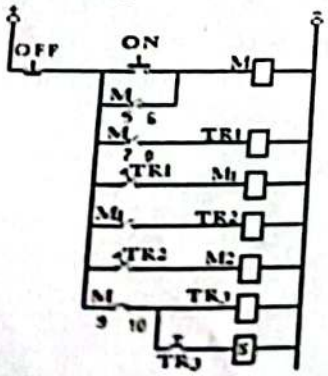
18. In the figure below, what is the correct statement?
a. if the load exceeds the rated value by 20%, the thermal relay will disconnect the motor within 10 Mins.
b. if the load exceeds the rated value by 40%, the thermal relay will disconnect the motor within 10 Mins.
c. through 10Mins, the motor may be loaded by 20% or 40%
d. none of these



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19. The function of the N/C contacts in control circuit below is:
a. to prevent short circuit in the control circuit b. to prevent short circuit in the power circuit
c. to open the starting-switch leg of the opposite rotation circuit for several seconds, while the fan decelerates to a halt. d. none of these

20. The circuit below is used for
a. starting of dc motor in three steps and dynamic braking b. starting of dc motor in one step
c. starting of dc motor in two steps and dynamic braking d. none of these



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1- D

2- A

3- B

4- C

5- D

6- D

7- C

8- B

9- A

10- B

11- C

12- D

13- A

14- B

15- C

16- C

17- D

18- A

19- B

20- C

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Q2 : (10 marks)

A constant speed drive has the following duty cycles

(I) load rising linearly from 200-500kw: 4 min

(II) uniform load of 400kw:2 min

(III) regenerative power returned to the supply reducing linearly from 400kw to 0:3 min

(IV) remains idle: 4min

Determine power rating of the motor assuming loss to be proportional to $(p0wer)^2$

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